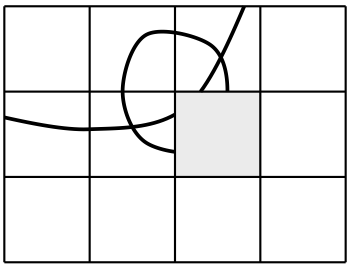
Preview: y2002_q01 (v1)

Find the missing square by its edges: lines must continue at the same places, not turned

Part A: Each question is worth 3 points

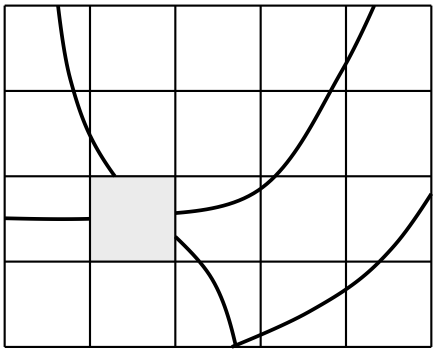
1. Mia drew some lines on squared paper. Then she cut out one square of her picture.



Which square was cut out of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

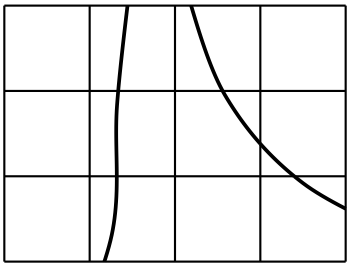
2. Leo painted a river and some roads on a map made of squares. One square of the map is missing.



Which square was cut out of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

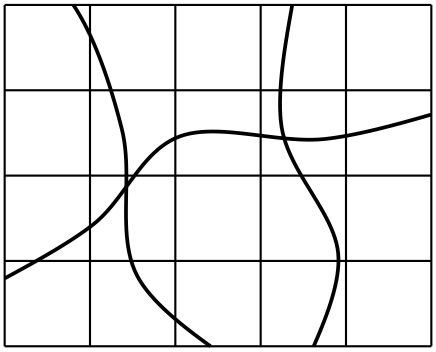
3. Mia drew some lines on squared paper. Here is the whole picture.



Which of these squares is **not** a piece of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

4. Leo painted a river and some roads on a map made of squares. Here is the whole picture.

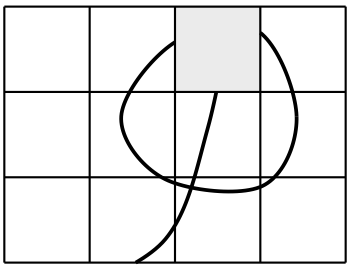


Which of these squares is **not** a piece of the picture?

A. ***B.*** ***C.*** ***D.*** ***E.***

Part B: Each question is worth 4 points

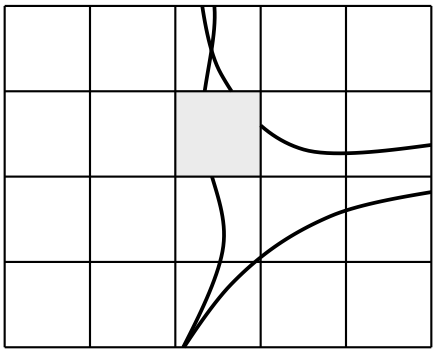
5. One square was cut out of this picture. Here it is five times, turned in different ways (and sometimes flipped over).



Which one fits back into the gap?

A. ***B.*** ***C.*** ***D.*** ***E.***

6. One square was cut out of this picture. Here it is five times, turned in different ways (and sometimes flipped over).

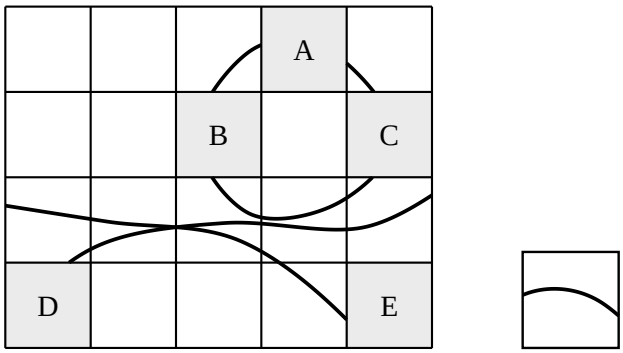


Which one fits back into the gap?

A. ***B.*** ***C.*** ***D.*** ***E.***

Part C: Each question is worth 5 points

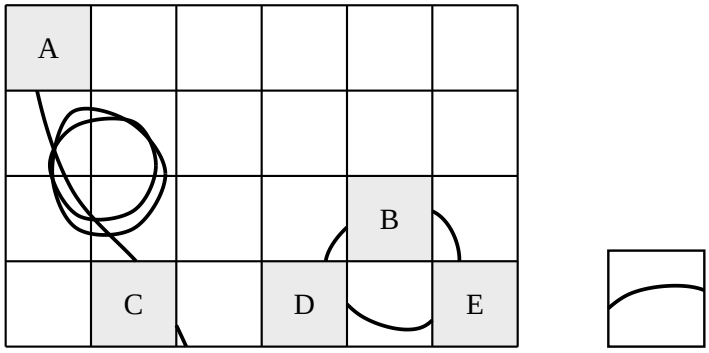
7. Mia drew some lines on squared paper. Five squares are missing, marked A to E. The square on the right is one of them (it is not turned).



Which gap does it fit into?

A. gap A ***B.*** gap B ***C.*** gap C ***D.*** gap D ***E.*** gap E

8. A cat played with balls of wool on squared paper. Five squares are missing, marked A to E. The square on the right is one of them (it is not turned).



Which gap does it fit into?

A. gap A ***B.*** gap B ***C.*** gap C ***D.*** gap D ***E.*** gap E

Answer Key

1. **A**	**2. E**	**3. A**	**4. A**	**5. B**	**6. B**
7. **A**	**8. B**				

Solutions

1. A — Look where the lines of the picture reach the grey gap: top: 2 lines, right: no line, bottom: no line, left: 2 lines. Only square A has its lines in exactly those places without being turned. The others are from another place, turned. [2002 Q1 · missing_square]

2. E — Look where the lines of the picture reach the grey gap: top: 1 line, right: 2 lines, bottom: no line, left: 1 line. Only square E has its lines in exactly those places without being turned. The others are from another place, mirrored and turned, mirrored and upside down. [2002 Q1 · missing_square]

3. A — Find each square in the picture by the places where its lines reach the edges. Squares B, C, D, E can be found; square A is a piece mirrored and turned, so it is not in the picture like that. [2002 Q1 · not_a_piece]

4. A — Find each square in the picture by the places where its lines reach the edges. Squares B, C, D, E can be found; square A is a piece mirrored and turned, so it is not in the picture like that. [2002 Q1 · not_a_piece]

5. B — Look where the lines of the picture reach the grey gap: top: 1 line, right: 1 line, bottom: 1 line, left: 1 line. Only square B has its lines in exactly those places in this position. The others are mirrored, mirrored and turned, turned, upside down. [2002 Q1 · turned_square]

6. B — Look where the lines of the picture reach the grey gap: top: 2 lines, right: 1 line, bottom: 1 line, left: no line. Only square B has its lines in exactly those places in this position. The others are mirrored, mirrored and turned, upside down. [2002 Q1 · turned_square]

7. A (gap A) — The square has top: no line, right: 1 line, bottom: no line, left: 1 line. Only gap A has lines reaching it at exactly those places. [2002 Q1 · which_gap]

8. B (gap B) — The square has top: no line, right: 1 line, bottom: no line, left: 1 line. Only gap B has lines reaching it at exactly those places. [2002 Q1 · which_gap]